

4 POINTS

① THEORY

MANY HAVE BEEN PENALIZED FOR NOT REPORTING THE INCREMENTAL VERSION OF THE ALGORITHM

② 4 POINTS

THEORY

SOME HAVE BEEN PENALIZED FOR NOT BEING CLEAR ABOUT WHAT THEY WERE LOOPING ON

③ 5 POINTS

THEORY

WE DISCUSSED A LOT ABOUT THIS.

'POOR' ANSWERED, EVEN IF PARTIALLY CORRECT, HAVE BEEN PENALIZED.

⑦ 3 POINTS

THEORY

⑨ 3 POINTS

YES, WE COULD USE OFF-POLICY AND IMPORTANCE SAMPLING.

IT WAS IMPORTANT HERE, FOR GETTING FULL POINTS, TO POINT OUT THAT WE HAVE COVERAGE IN THIS SETTING

TARGET POLICY μ

BEHAVIOUR POLICY π

$\Rightarrow$

COVERAGE SINCE

IF $\mu(a|s) > 0$

$\Rightarrow \pi(a|s) > 0$

④ 4 POINTS

STARTING $V_{\pi} = (0, 0.5, 0.5, 0.5, 0)$ $\gamma=1$ $\lambda=0.5$

EPISODE #01

$$V_{\pi}(B) = 0.5 + 0.5(1 - 0.5) = 0.75$$

$$V_{\pi}(C) = 0.5 + 0.5(1 - 0.5) = 0.75$$

$$\Rightarrow V_{\pi} = (0, 0.5, 0.75, 0.75, 0)$$

EPISODE #02

$$V_{\pi}(A) = 0.5 + 0.5(0 - 0.5) = 0.25$$

$$V_{\pi}(B) = 0.75 + 0.5(0 - 0.75) = 0.375$$

$$V_{\pi}(C) = 0.75 + 0.5(0 - 0.75) = 0.375$$

$$\Rightarrow V_{\pi} = (0, 0.25, 0.375, 0.375, 0)$$

⑤ 4 POINTS

STARTING $V_{\pi} = (0, 0.5, 0.5, 0.5, 0)$ $\gamma=1$ $\lambda=0.5$

EPISODE #01

$$V_{\pi}(B) = 0.5 + 0.5(0 + 0.5 - 0.5) = 0.5$$

$$V_{\pi}(C) = 0.5 + 0.5(1 + 0 - 0.5) = 0.75$$

$$\Rightarrow V_{\pi} = (0, 0.5, 0.5, 0.75, 0)$$

EPISODE #02

$$V_{\pi}(B) = 0.5 + 0.5(0 + 0.75 - 0.5) = 0.625$$

$$V_{\pi}(C) = 0.75 + 0.5(0 + 0.625 - 0.75) = 0.6875$$

$$V_{\pi}(B) = 0.625 + 0.5(0 + 0.5 - 0.625) = 0.5625$$

$$V_{\pi}(A) = 0.5 + 0.5(0 + 0 - 0.5) = 0.25$$

$$\Rightarrow V_{\pi} = (0, 0.25, 0.5625, 0.6875, 0)$$

6 4 POINTS

- THE TRUE V_{π} IS CLEARLY $V_{\pi} = (0, 0.25, 0.5, 0.75, 0)$
(SOME USED THE FULL KNOWLEDGE OF THE MDD AND BELLMAN EQUATIONS TO COMPUTE IT, WHICH WAS NOT NECESSARY)

	TRUE	MC	TD(0)
$V_{\pi}(A)$	0.25	0.25	0.25
$V_{\pi}(B)$	0.5	0.375	0.5625
$V_{\pi}(C)$	0.75	0.375	0.6875

$\left. \begin{array}{l} \text{MC} \\ \text{TD(0)} \end{array} \right\} \text{SQUARED ERROR!}$

$(0.5 - 0.375)^2 + (0.75 - 0.375)^2 =$
 ≈ 0.156

$(0.5 - 0.5625)^2 + (0.75 - 0.6875)^2 =$
 ≈ 0.008

- TD(0) DID A BETTER JOB.
- GENERIC ANSWERS (TD(0) BETTER THAN MC) HAVE NOT BEEN AWARDED ANY POINTS
- ANSWERS THAT DID NOT PROVIDE THE TRUE V_{π} , BUT REALIZED THAT WITH

TD(0): $V_{\pi}(B) < V_{\pi}(C)$

WHILE WITH:

MC: $V_{\pi}(B) = V_{\pi}(C)$

AND CONCLUDING THAT TD(0) ACHIEVED A BETTER ESTIMATION WERE GIVEN PARTIAL POINTS

⑧ [OPTIONAL] 2 POINTS

THIS WAS WRITTEN IN A WAY THAT WAS A LITTLE MISLEADING, SO IT WAS CONSIDERED OPTIONAL

- THE KNOWLEDGE OF Q WAS NECESSARY FOR THIS, OTHERWISE WE ARE NOT IN FULL RL SETTINGS (WHICH IS OBVIOUS SINCE WE APPLY MC AND TD(0))
- THOSE WHO RESPONDED CONSIDERING $P(S, A)$ KNOW BUT PROVIDING THE RIGHT ANSWER

$$\pi_{TD(0)}(-\rightarrow | S_j) \quad \forall S_j$$

$$\pi_{MC}(-\rightarrow | S_j) \quad \forall S_j$$

STILL GOT 1 POINT

⑩ [OPTIONAL] 2 POINTS

- REALLY FEW PEOPLE ANSWERED THIS, IN BOTH CASES CONSIDERING TD(0) OR MC.
- THE IMPORTANT THING WAS TO CONSIDER

TARGET POLICY μ

BEHAVIOUR POLICY π

$\Rightarrow$ IMPORTANCE SAMPLING

$$\pi \frac{\mu(A|S)}{\pi(A|S)}$$

MOVIE CARO: EPISODE #01

$$w_1 = \frac{\mu(\rightarrow | C)}{\pi(\rightarrow | B)} = \frac{0.75}{0.5} = 1.5$$

$$w_0 = \frac{\mu(\rightarrow | B)}{\pi(\rightarrow | B)} \Rightarrow w_1 = \frac{0.75}{0.5} \cdot 1.5 = 2.25$$

$$V_\pi(C) = 0.5 + 0.5 \cdot 1.5 (1 - 0.5)$$

$$= 0.5 + 0.375 = 0.875$$

$$V_\pi(B) = 0.5 + 0.5 \cdot 2.25 (1 - 0.5)$$

$$= 0.5 + 0.5625 = 1.0625$$

OTHER APPROACHES
(FOR EXAMPLE, NON INCREMENTAL)
WERE POSITIVELY
EVALUATED

EPISODE #02

$$w_3 = \frac{\mu(\leftarrow | A)}{\pi(\leftarrow | A)} = \frac{0.25}{0.5} = 0.5$$

$$w_2 = \frac{\mu(\leftarrow | B)}{\pi(\leftarrow | B)} \cdot w_3 = \frac{0.25}{0.5} \cdot 0.5 = 0.25$$

$$w_1 = \frac{\mu(\leftarrow | C)}{\pi(\leftarrow | C)} \cdot w_2 = \frac{0.25}{0.5} \cdot 0.25 = 0.125$$

$$w_0 = \frac{\mu(\rightarrow | B)}{\pi(\rightarrow | B)} \cdot w_1 = \frac{0.75}{0.5} \cdot 0.125 = 0.1875$$

FIRST VISIT

$$V_\pi(A) = 0.5 + 0.5 \cdot 0.5 (0 - 0.5) = 0.5 - 0.125 = 0.375$$

$$V_\pi(B) = 1.0625 + 0.5 \cdot 0.1875 (0 - 1.0625) \approx 1.0625 - 0.0996$$

$$\approx 0.9629$$

$$V_\pi(C) = 0.875 + 0.5 \cdot 0.125 (0 - 0.875) \approx 0.875 - 0.0547$$

$$\approx 0.8203$$